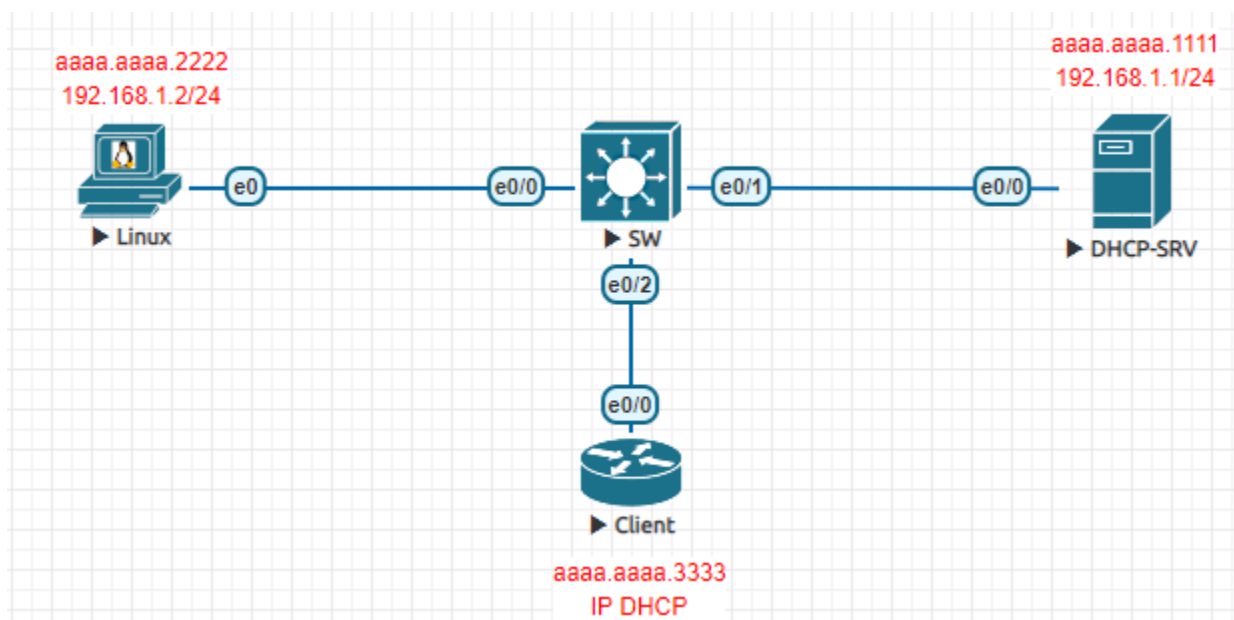
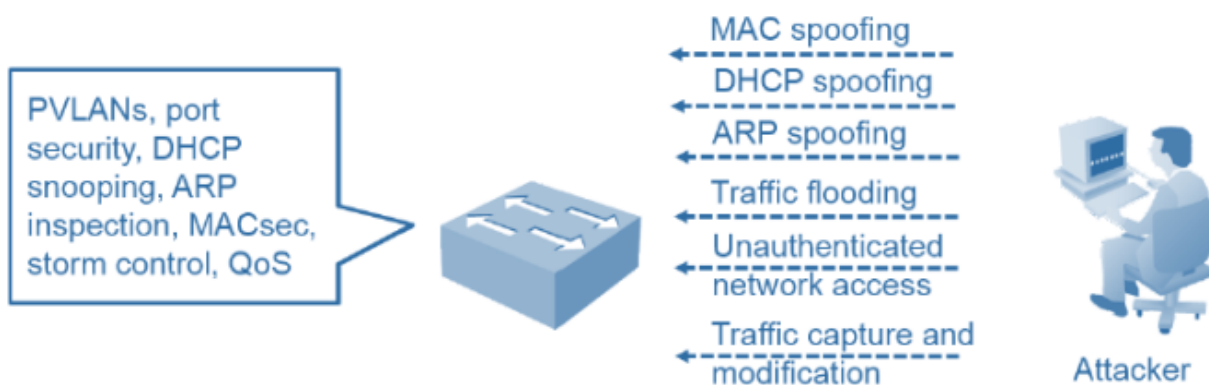


## Layer 2 Attacks:

- o Most end-user devices are connected to the network via Layer 2 access switches.
- o End devices such as Computers, Printers, IP Phones, fix machine & other hosts etc.
- o As a result, mostly, Access Layer Cisco switches can present a network security risk.
- o The Cisco Switches are more subject to attack from malicious internal users etc.
- o Cisco Switches provide many security features to block internal attacks of users.
- o The Security of Layer 2 is aims to mitigate Man in The Middle Attacks (MITM).
- o ARP Poisoning, MAC Attack, Spoof Attack and DHCP snooping are layer-2 attacks.



### DHCP SERVER Configuration

```
DHCP-SRV(config)#interface ethernet0/0
DHCP-SRV(config-if)#ip address 192.168.1.1 255.255.255.0
DHCP-SRV(config-if)#mac-address aaaa.aaaa.1111
DHCP-SRV(config-if)#no shutdown

DHCP-SRV (config)#ip dhcp excluded-address 192.168.1.1 192.168.1.100
DHCP-SRV (config)#ip dhcp pool TEST
DHCP-SRV (dhcp-config)#network 192.168.1.0 /24
DHCP-SRV # show ip dhcp binding
```

### DHCP Client Configuration

```
Router(config)#hostname client
client(config)#interface ethernet0/0
client(config-if)#mac-address aaaa.aaaa.3333

client(config)#interface ethernet0/0
Client(config-if)#no ip address
Client(config-if)#ip address dhcp
Client# show ip int br
```

### Kali Linux Configuration

```
root@kali:~# ifconfig eth0 192.168.1.2 netmask 255.255.255.0
root@kali:~# macchanger -m aa:aa:aa:aa:22:22
```

### Switch Configuration

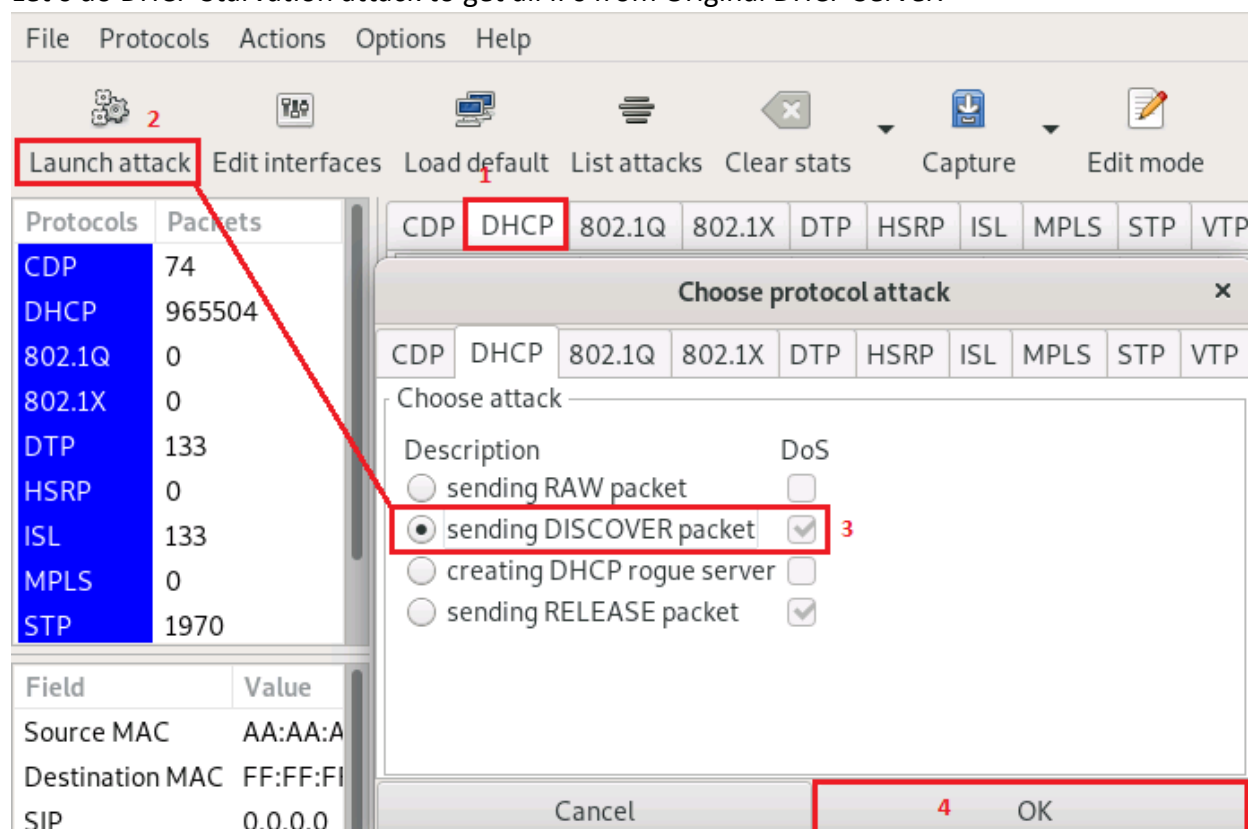
```
SW(config)#interface vlan 1
SW(config-if)#ip address 192.168.1.4 255.255.255.0
SW(config-if)#mac-address aaaa.aaaa.4444
SW(config-if)#no shutdown
```

```
client(config-if)#do show ip int br
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 192.168.1.104 YES DHCP up up
Ethernet0/1 unassigned YES unset administratively down down
Ethernet0/2 unassigned YES unset administratively down down
Ethernet0/3 unassigned YES unset administratively down down
```

```
DHCP-SRV(dhcp-config)#do show ip dhcp bin
Bindings from all pools not associated with VRF:
IP address Client-ID/ Lease expiration Type
Hardware address/
User name
192.168.1.104 0063.6973.636f.2d61. Aug 20 2020 01:45 PM Automatic
6161.612e.6161.6161.
2e33.3333.332d.4574.
302f.30
```

## DHCP Starvation Attack:

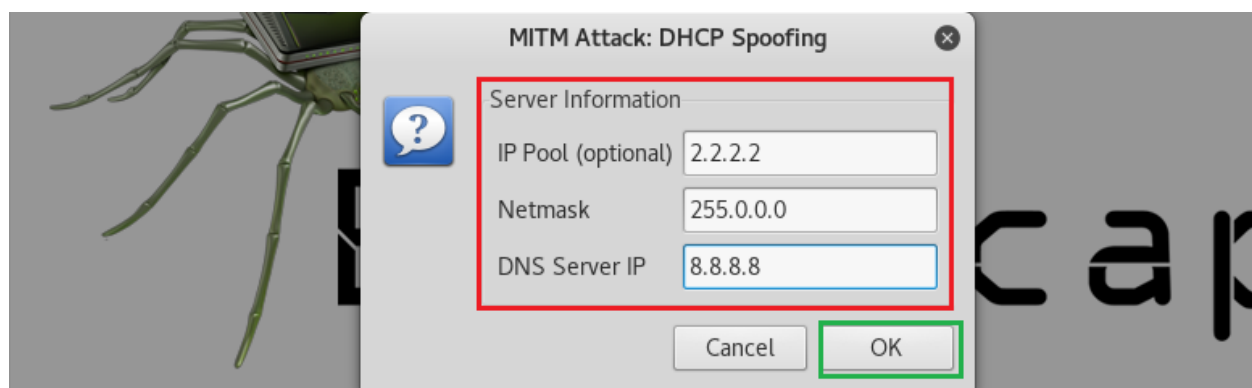
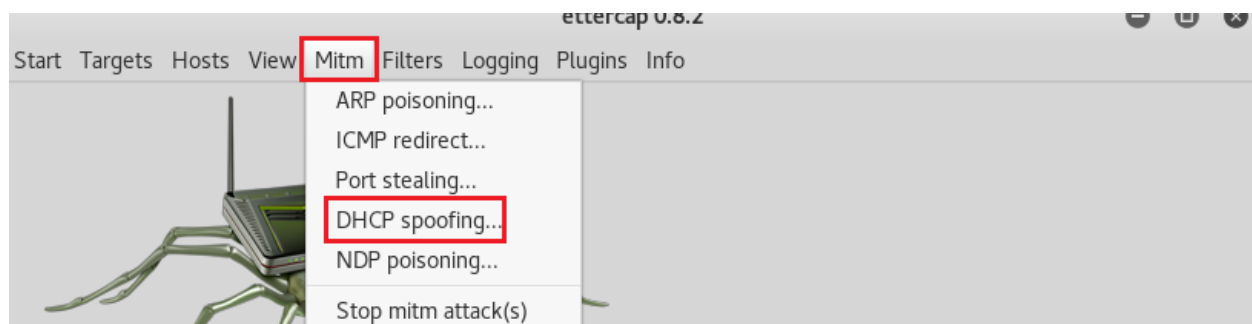
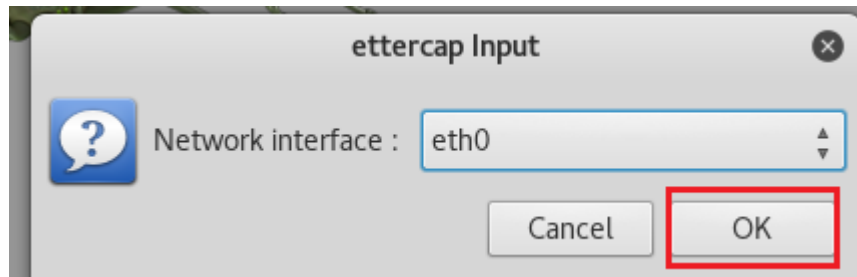
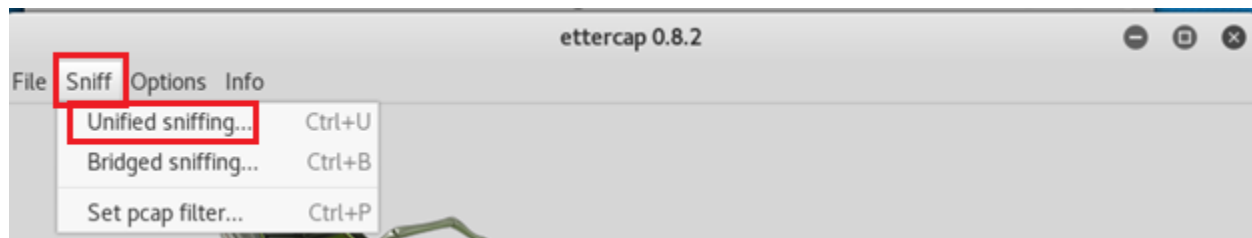
Let's do DHCP Starvation attack to get all IPs from Original DHCP Server.



All the IPs are taken by attacker using Starvation there is no more IPs for clients.

```
DHCP-SRV(dhcp-config)#do show ip dhcp bin
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
Hardware address/
User name
192.168.1.101    e0c1.4200.22a9   Aug 19 2020 01:48 PM    Automatic
192.168.1.102    0100.5000.0001.00 Aug 20 2020 01:18 PM    Automatic
192.168.1.103    64a7.ac20.10b9   Aug 19 2020 01:48 PM    Automatic
192.168.1.104    0063.6973.636f.2d61. Aug 20 2020 01:45 PM    Automatic
                  6161.612e.6161.6161.
                  2e33.3333.332d.4574.
                  302f.30
192.168.1.105    7a12.7e20.c516   Aug 19 2020 01:48 PM    Automatic
192.168.1.106    0c66.9008.8823   Aug 19 2020 01:48 PM    Automatic
192.168.1.107    b61f.6c6e.aadf   Aug 19 2020 01:48 PM    Automatic
192.168.1.108    5ae6.ed6a.bb92   Aug 19 2020 01:48 PM    Automatic
192.168.1.109    7045.ad0d.6020   Aug 19 2020 01:48 PM    Automatic
192.168.1.110    2e20.5d30.7f83   Aug 19 2020 01:48 PM    Automatic
192.168.1.111    2628.b805.bc75   Aug 19 2020 01:48 PM    Automatic
192.168.1.112    16ac.0f22.e6ea   Aug 19 2020 01:48 PM    Automatic
192.168.1.113    588d.410f.4e8f   Aug 19 2020 01:48 PM    Automatic
```

## Rogue DHCP Server:



```
Client(config-if)#do show ip int br
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0    2.2.2.2         YES DHCP    up          up
FastEthernet1/0    unassigned      YES NVRAM   administratively down down
```

## DHCP Snooping:

- o DHCP snooping is a security feature acts like a firewall between trusted & untrusted.
- o DHCP snooping is like a firewall between untrusted hosts and trusted DHCP servers.
- o DHCP snooping feature use trusted source to reply the DHCP offer message.
- o DHCP snooping will drop DHCP messages from a DHCP server that is not trusted.
- o DHCP snooping rate-limits DHCP traffic from trusted and untrusted sources.
- o DHCP snooping keep binding database, which is untrusted hosts with leased IPs.
- o DHCP snooping binding database validate subsequent requests from untrusted hosts.
- o DHCP snooping Can be enable to disabled per Virtual Local Area Network basis.
- o DHCP snooping feature is inactive on all Virtual Local Area Network by default,
- o DHCP snooping device insert **DHCP option No 82 Relay Agent Information Option**.
- o DHCP Snooping is used to prevent a Man-In-The Middle attack on the network.
- o DHCP Snooping uses a mechanism and technique of trusted and untrusted sources.
- o DHCP Snooping is L 2 security switch feature to blocks unauthorized DHCP servers.
- o DHCP snooping feature identifies Switch Ports as "trusted" and "untrusted".
- o DHCP Untrusted interfaces where clients are connected cannot source DHCP messages.
- o DHCP trusted interfaces where Server are connected source all types of DHCP messages.
- o DHCP Snooping provides protection from DHCP starvation & Rogue DHCP Server attacks.

| DHCP Snooping Configuration SW                                        |  |
|-----------------------------------------------------------------------|--|
| SW(config)#ip dhcp snooping                                           |  |
| SW(config)#ip dhcp snooping vlan 1                                    |  |
| SW(config)#interface Ethernet 0/1                                     |  |
| SW(config-if)#ip dhcp snooping trust                                  |  |
| SW(config-if)# no ip dhcp snooping information option                 |  |
| SW(config)#interface Ethernet 0/1                                     |  |
| SW(config-if)# ip dhcp snooping limit rate 100                        |  |
| SW(config)#interface range Ethernet 0/0, Ethernet 0/2                 |  |
| SW(config-if)# ip dhcp snooping limit rate 20 (20 packets per second) |  |
| SW# show ip dhcp snooping binding                                     |  |
| SW# show ip dhcp snooping                                             |  |
| SW# show ip dhcp snooping statistics                                  |  |

```
SW#show ip dhcp snooping binding
MacAddress      IpAddress      Lease(sec)  Type           VLAN  Interface
-----
AA:AA:AA:AA:33:33  192.168.1.105  86380      dhcp-snooping  1     Ethernet0/2
Total number of bindings: 1
```

```
SW#show ip dhcp snooping binding
MacAddress      IpAddress      Lease(sec)  Type           VLAN  Interface
-----
AA:AA:AA:AA:33:33  192.168.1.105  86380      dhcp-snooping  1     Ethernet0/2
Total number of bindings: 1
```

```
SW#show ip dhcp snooping
Switch DHCP snooping is enabled
Switch DHCP gleaning is disabled
DHCP snooping is configured on following VLANs:
1
DHCP snooping is operational on following VLANs:
1
DHCP snooping is configured on the following L3 Interfaces:
```

**Insertion of option 82 is disabled**

```
circuit-id default format: vlan-mod-port
remote-id: aabb.cc00.2000 (MAC)
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:
```

| Interface           | Trusted | Allow option | Rate limit (pps) |
|---------------------|---------|--------------|------------------|
| Ethernet0/0         | no      | no           | 20               |
| Custom circuit-ids: |         |              |                  |
| Ethernet0/1         | yes     | yes          | 100              |
| Custom circuit-ids: |         |              |                  |
| Ethernet0/2         | no      | no           | 20               |
| Interface           | Trusted | Allow option | Rate limit (pps) |
| Custom circuit-ids: |         |              |                  |